

CML Project

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1 Question 1

We started by constructing a causal diagram seen in Figure 1 to find possible "backdoors". First, the causal link between e401k and net financial assets is drawn, followed by the effects of income. Naturally, income will affect the financial assets, but higher paying jobs are also more likely to pay out a pension scheme. Lastly, the unobserved saving preference is added, this will affect the income of a person and might further affect financial assets in ways we cannot observe. This leads to two possible backdoors in our diagram; luckily, adjusting for income closes both these doors. So, we can conclude that given the situation sketched in the assignment, e401k is exchangeable conditional on income.

2 Question 2

The naive estimation $E(Y|A=1) - E(Y|A=0)$ yields an average treatment effect of \$19559.34. This doesn't account for possible confounders so we used Augmented Inverse Probability Weighting (AIPW) and Targeted Maximum Likelihood Estimation (TMLE). These doubly robust methods are used to account for a range of possible confounders like age, marital status and years of education. We used the functions provided to us via the course and chose `SL.glm`, `SL.step`, `SL.interaction`, `SL.glmnet` and `SL.ranger`. We considered `SL.randomForest`, but due to computational difficulties we opted to drop it.

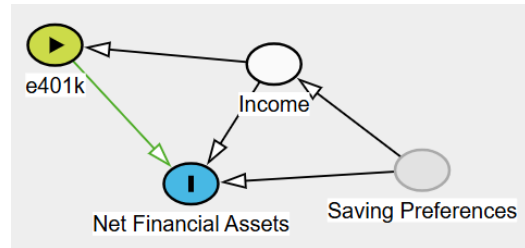


Figure 1: Causal diagram involving both Income and saving preference

The results can be found in the table below: The large difference in estimates shows that

Method	Estimate	Standard Error
Naive	19559.34	1923.259
AIPW	7090.647	1289.329
TMLE	7094.58	1289.262

we were justified in controlling for the chosen variables. We still found a sizeable average



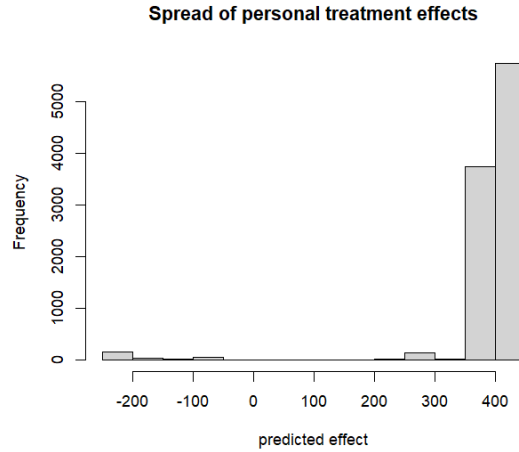


Figure 2: A histogram of the predicted personal treatment effects

treatment effect and both AIPW and TMLE showed similar estimates. This tells us that our methods are rather robust and that the conclusions are probably representative.

3 Question 3

For this question, we primarily used the GLM SuperLearner, and 10-fold cross-validation. Upon calculating the estimator for θ , $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n (A_i - E[A_i|L_i])(Y_i - E[Y_i|L_i])$, we obtained 1042.1, indicating it is likely significant. The obtained standard error was 155.9221, with which we calculated a confidence interval for the estimator: [736.4931, 1347.708]. As 0 is not included in this confidence interval, we can assume that eligibility for a 401(k) does, in fact, affect net financial assets Y. The obtained p-value is of 2.333533e-11, once again indicating that the effect is significant.



4 Question 4

Implementation wise, this part of the assignment was very straightforward. The package "rlearner" allowed us to simply calculate the needed predicted values. However, executing the code was the difficult part. `rlasso()` proved to be an inadequate method as all the predicted values were the same. We then tried using `rboost()` which took a really long time.

After using `doParallel`, we cut the computational time by a lot and obtained the predicted values. These were still very similar, but there were some differences. We see in Figure 2 that the effect is mostly concentrated around the 400 mark, but there are some other hotspots around 300 and -200



All of the three variables of interest show similar behaviour; there is a strong pattern of diminishing returns. Overall, there is positive association between the treatment effect and age, income, and years of education. However, this effect levels off very strongly from a certain threshold.

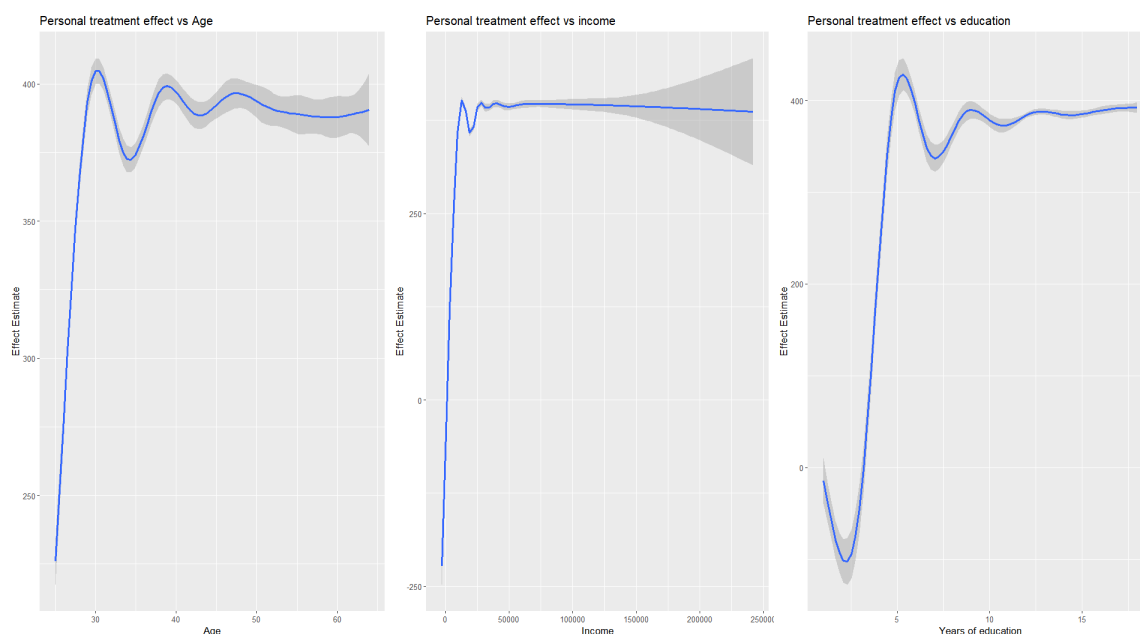


Figure 3: Marginal treatment effects of Age (left), Income (middle) and Education (right)

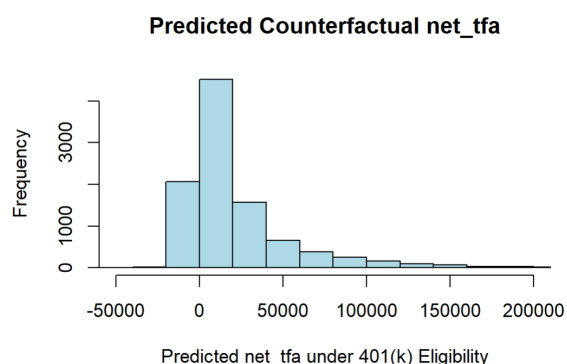


Figure 4: A histogram of the predicted counterfactual net_tfa under 401(k) eligibility

5 Question 5

For this question, we first estimated $\hat{Y}_1$ and the propensity scores using the given μ_hat and π_hat functions, and calculated the pseudo-outcomes. Using ten-fold sampling splitting, we obtained $\tilde{Y}^1$, which as shown plotted above on a histogram.

We calculated $E[(Y^1 - \tilde{Y}^1)^2]$ using our results for $\tilde{Y}^1$; this gave us a value of 3778277754, with a confidence interval of [2102686179, 5453869328].

6 Comments

We split this task equally; Joren did questions 1, 2 and 4, while Leire did questions 3 and 5.